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Research Assignment 1.

Section A: Data Fundamentals

1. A database is an Organized Collection of data that is Stored electronically and can be easily accessed, Managed and updated. Database are essential because they allow bussinesses and Systems to Store large amounts of information efficiently and retrieve it quickly when needed.

Examples:
- A Bank Storing Customer account details and transactions
- A Hospital Storing patient records and Medical history

2. A data Warehouse is a system Used to Store large amounts of historical data from diffrent Sources for analysis and reporting.

Difference from database
- Database: Used for daily Operations (Transactions)
- Data Warehouse: Used for analysis and decision-Making.

Why Companies Use it:
- To analyze trends, improve decision-making and generate reports.

3. What is a data lake?
- A data lake is a Storage System that holds raw data in its Original formart (Structured and unstructured).

Difference from data warehouse:

Data lake: raw, unprocorsed data.
Data Warehouse: Cleaned and Structured data.

When Used:
- When Companies wants to Store large Volumes of data for future analysis or AI.

5. Types of data

1. Structured data: Organised in tables (Excel, Spreadsh
2. Semi-Structured data: Has some Structure (JSon, XML
3 Unstructured data: No fixed format (Videos, emails)

6. Data Modeling is the process of designing how data is stored and related in a System.

Importance
- Improves efficiency
- Ensures accuracy
- Helps in System design.

Types
- Conceptual model
- Logical model.

7. A Database Managment System (DBMS) is a Software used to Create and Manage databases.

Examples.
~~MSSO~~ MySQL - Used for Web application.
PostgreSQL - Advanced data handling.
Oracle - large enterprice Systems
MongoDB - Handles Unstructured data.

9. What is ETL?
ETL Mears / Stands for:

E : ~~Get dat~~ Extract : Get data from Sources
T : Transform : Clear and format data
L : Load : Store data into a System.

Ensures data is accurate and ready for analyisis.

10. Data Storage formats.

- Csv : Simple table format
- Json : Used in API's
- Parquet : Efficient for Big data
- Avro : Used in data Streaming

Section B: AI and Machine Learning

11. Artifical Intelligence / AI

AI is the ability of Machines to promote / perform tasks that require human Intelligence
- Narrow AI : performs Specific tasks (voice assitance)
- General AI : Can perform any human task (still theaical).

12. Machine Learning (ML):
- ML allows Computers to learn from data instead of being programmed.

Type:
- Supervised learning.
- Unsupervised learing.
- Reinforcement learning.

13. Deep learning
- A Subset of ML that Uses neural network to process Complex data.

Neural Networks:
- Systems inspired by the human brain that learn patterns.

14. NLP:
Natural Language Processing allows machines to Understand human language

example:
Chatbots ; Voice assistants , Spell Check

15. Large Language Models (LLMs):
LLMs are advanced AI Models trained on large text datasets.

examples:
ChatGPT - Conversation , Claude - Writing and analysis.

16. Generative AI
AI that Create new Content (text, images, etc)
Tools:
ChatGPT - Text, DALL-E - images, Midjourney - artwork.

18. Computer Vision:
Allows Machines to interpret images

examples:
Face recognition, Self driving Cars.

17. Training Dataset:
- Data Used to train AI models

Importance:
- Bad or Biased data leads to incorrect results.

Section C: API, and Tools

19. API:
An API allows systems to Communicate

Analogy:
Like a Waiter taking your order to the kitchen

20. API key:
A Code Used a access an API Securly
Risk if exposed: Unauthorized access and data misuse
21. Rest API:

REST = Representational State Transfer
Methods :

Get - Retrieve data
Post - Send data
Put - Update data
delete - Remove data

22. Json :
- A lightweight data format.
example :

Json
{
"name" : "John",
"age" : 25
}

23. Prompt Engineering :
Write / Writing Clear instructions for AI

Example :
- Poor : "Write Something about data"
- Better : "Explain data analytics in 100 words With Examples"

24. MAP Function :

- Applies a function to each item in a list.

Example (Python):

Python
numbers = [1,2,3]
result = list(map(lambda x: x*3, numbers))

25. Cloud Computing

- IaaS: Infrastructure (eg AWS EC2)
- PaaS: Platform (eg Google App Engine)
- SaaS: Software (Gmail)

26. Version Control
Tracks Changes in Code.

GIT: Most popular tool
- Commit: Saves Changes
- Branch: Separate Version of Code

27. Jupyter Notebook:

A tool for writing and running Code

Language: Python

Why popular
Interactive and easy for analysis

Libraries

Pandas - data analysis
Num Py - Calculations
Matplotlib - Visualization
SCikit-learn - Machine learning.

Section D : Data Ethics

29. Data Privacy

Protecting personal data.

GDPR :
A law in the EU that protects user data.

30. Data Ethics

Examples
- Selling personal data without consent
- Biased AI decisions
- Surveillance Misuse

31. Data Governance

Managing data quality and Security

Componets :
Polices
Data quality
Security
Compliance.

32. Bias in AI:
When AI Produces Unfair results.

Example.
- Hiring Systems favoring Certain groups.

33.

Role	Focus	Tools
Data Analyst	Analyze data	Excel, SQL
Data Scientist	Build Models	Python, ML
Data Engineer	Build Pipelines	SQL, ETL
AI Engineer	Build AI System	Python, AI Frameworks

34. Business Intelligence (BI):
- Tools that help analyze data.
Example:
- PoweBI
- Tableau

Section E: Reflection

35. Most interesting:
AI and Machine Learning because they can automate tasks and make smart decisions.

36. Most difficult:
Deep Learning - Used youtube to Understand it

37. Future Impact
Data and AI will help automate jobs, improve decisions-making and Create new Career oppotunites.